MEDIUM · ALGEBRA · ANSWER KEY

Linear equations in one variable

03

7 answers · Rationale-rich · Teach from doc

Q1 D

D. -10

Choice D is correct. Multiplying the given equation by 2 on each side yields $2\left(4x-\frac{1}{2}\right)=2(-5)$. Applying the distributive property, this equation can be rewritten as $2(4x)-2\left(\frac{1}{2}\right)=2(-5)$, or $8x-1=-10$.

Choices A, B, and C are incorrect and may result from calculation errors in solving the given equation for x and then substituting that value of x in the expression $8x-1$.

Q2 60

The correct answer is 60. Subtracting t from both sides of the given equation yields $2(3t - 10) = 40 + 3t$. Applying the distributive property to the left-hand side of this equation yields $6t - 20 = 40 + 3t$. Adding 20 to both sides of this equation yields $6t = 60 + 3t$. Subtracting $3t$ from both sides of this equation yields $3t = 60$. Therefore, the value of $3t$ is 60.

Q3 294

The correct answer is 294. Subtracting 18 from each side of the given equation yields $\frac{6}{7}p = 36$. Multiplying each side of this equation by $\frac{7}{6}$ yields $p = 42$. Multiplying each side of this equation by 7 yields $7p = 294$. Therefore, the value of $7p$ is 294.

Q4

C

C. 79

Choice C is correct. For the given equation, $x + 5$ is a factor of both terms on the left-hand side. Therefore, the given equation can be rewritten as $\frac{1}{4} - \frac{1}{3}x + 5 = -7$, or $\frac{3}{12} - \frac{4}{12}x + 5 = -7$, which is equivalent to $-\frac{1}{12}x + 5 = -7$. Multiplying both sides of this equation by -12 yields $x + 5 = 84$. Subtracting 5 from both sides of this equation yields $x = 79$.

Choice A is incorrect. This is the value of x for which the left-hand side of the given equation equals $\frac{7}{12}$, not -7.

Choice B is incorrect. This is the value of x for which the left-hand side of the given equation equals 0, not -7.

Choice D is incorrect. This is the value of x for which the left-hand side of the given equation equals $-\frac{209}{12}$, not -7.

Q5

22

The correct answer is 22.4. If the height of the plant increased by an average of 1.20 centimeters per day for 12 days, then its total growth over the 12 days was $(1.20)(12) = 14.4$ centimeters. The plant was 36.8 centimeters tall after 12 days, so at the beginning of the study its height was $36.8 - 14.4 = 22.4$ centimeters. Note that 22.4 and 112/5 are examples of ways to enter a correct answer.

Alternate approach: The equation $36.8 = 12(1.20) + h$ can be used to represent this situation, where h is the height of the plant, in centimeters, at the beginning of the study. Solving this equation for h yields 22.4 centimeters.

Q6

D

D. 16

Choice D is correct. The value of $4 - 3x$ can be found by isolating this expression in the given equation. Subtracting 2 from both sides of the given equation yields $94 - 3x = 84 - 3x + 16$. Subtracting $84 - 3x$ from both sides of this equation yields $94 - 3x - 84 - 3x = 16$, which gives $14 - 3x = 16$, or $4 - 3x = 16$. Therefore, the value of $4 - 3x$ is 16.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not $4 - 3x$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7

37

The correct answer is 37. It's given that the museum earns revenue of $\$14$ for each tablet rented for the day. It's also given that on Wednesday, the museum earned $\$406$ in profit from renting tablets after paying daily expenses of $\$112$. Let x represent the number of tablets the museum rented on Wednesday. It follows that the total revenue can be represented by the expression $14x$. Because $\text{profit} = \text{total revenue} - \text{total expenses}$, the equation $406 = 14x - 112$ represents this situation. Adding 112 to both sides of this equation yields $14x = 518$. Dividing both sides of this equation by 14 yields $x = 37$. Therefore, the museum rented 37 tablets on Wednesday.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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